RL-Based Prompt Optimization for the Segment Anything Model

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Introduction

The Segment Anything Model (SAM) [1] enables interactive segmentation by accepting point prompts, boxes, or masks as input and producing segmentation masks as output. However, SAM's output quality is highly sensitive to the placement of prompt points, and its mask decoder commits to a single hypothesis early in its computation, making it difficult to correct once an initial interpretation is established. These issues are manageable with a human in the loop, but become critical in automated or one-shot segmentation settings where a reference image and mask must be used to segment the same object class in novel query images without manual intervention.

This project investigates four approaches to automating SAM prompt optimization. We first implemented a graph-based Deep Q-Network derived from the Plug-and-Play PPO framework [2], which failed due to lossy state encoding and proxy reward misalignment. We then developed a PolicyTransformer (27M parameters) trained via behavioral cloning followed by PPO fine-tuning, achieving 0.549 Dice before policy collapse during RL. A third approach extended this architecture with sub-mask decomposition actions (37M parameters), but the policy never discovered the decomposition strategy. Our fourth and most successful approach deploys Qwen3-VL-8B [3] as a zero-shot VLM agent, achieving 0.776 mean Dice with only 1.3 clicks on a 15-class FSS-1000 [4] ablation, outperforming all trained policies without any task-specific training.

Related Work

Our work builds on several lines of research. Foundation models for segmentation, particularly SAM [1] and its successors, provide the base segmentation engine. Prior work on RL-based click optimization, including SeedNet [5] and AIES [6], demonstrates that reinforcement learning can learn sequential click placement strategies for interactive segmentation, though these methods optimize proxy rewards rather than direct segmentation quality. The Plug-and-Play PPO framework [2] is the most direct predecessor, framing prompt selection as graph-based Q-learning over DINOv2 [7] feature spaces. Our project differentiates itself from these prior approaches in three ways. First, we replace proxy rewards with direct SAM Dice feedback during training. Second, we introduce a novel sub-mask decomposition action space that allows the agent to address SAM's hypothesis commitment problem by segmenting complex objects in parts. Third, we are the first to deploy a vision-language model as a zero-shot interactive segmentation policy, demonstrating that chain-of-thought visual reasoning can outperform all RL-trained policies without any task-specific optimization.

Approach

We explored four progressively refined architectures, all evaluated on the FSS-1000 benchmark [4]. The first approach (Q-learning) adopted the Plug-and-Play PPO framework's dual-space heterogeneous graph with DINOv2 features, using a Deep Q-Network to select optimal prompt subsets. The second approach (V1 PolicyTransformer) replaced the lossy graph representation with full $37 \times 37 \times 1024$ DINOv2 spatial features and trained a 27M parameter transformer via behavioral cloning on oracle demonstrations followed by PPO fine-tuning, using direct SAM Dice as the reward signal. The third approach (V2 SubMaskPolicyTransformer, 37M parameters) extended V1 with a five-action space including sub-mask decomposition and abandonment actions, trained via both GRPO and PPO. Each of these RL approaches revealed specific failure modes: the Q-learning proxy reward was anti-correlated with true segmentation quality, V1 suffered catastrophic policy collapse during PPO fine-tuning, and V2 never discovered the decomposition strategy despite its availability.

The fourth approach (V3) replaced learned policies entirely with Qwen3-VL-8B-Instruct [3] operating zero-shot. The VLM receives a four-panel visual state (reference image, query image, current mask overlay, and error heatmap) and follows a three-phase decision loop: decide on an action, verify the proposed click location, and review the result after execution. This architecture enables post-click review and undo, sub-mask decomposition through explicit reasoning, and early stopping when the mask is already sufficient. No task-specific fine-tuning or in-context demonstrations are used.

Experimental Results

Table 1 presents a controlled cross-architecture comparison on 15 FSS-1000 validation classes. V1 (best PPO checkpoint before collapse) achieves 0.411 mean Dice with 8.7 clicks. V2 (best GRPO checkpoint) improves to 0.610 Dice with 6.0 clicks through its richer action space and additional parameters. V3 (Qwen3-VL-8B, zero-shot) achieves 0.776 mean Dice with only 1.3 clicks, outperforming both trained policies on every metric including stop success rate (67% vs. 32% vs. 14%) and high-quality segmentation rate (60% of episodes above 0.85 Dice). The best single V3 prompt iteration reached 0.853 Dice with 1.4 clicks on a 5-image subset.

Metric	**V1 (RL Policy)**	**V2 (RL + SubMask)**	**V3 (VLM)**
Mean Dice	0.411	0.610	**0.776**
Mean IoU	0.342	0.521	**0.681**
Stop Success Rate	14%	32%	**67%**
Dice ≥ 0.85	4/15 (27%)	8/15 (53%)	**9/15 (60%)**
Mean Clicks	8.7	6.0	**1.3**

Table 1: Cross-architecture comparison on 15 FSS-1000 validation classes. V1: PolicyTransformer (BC + PPO). V2: SubMaskPolicyTransformer (BC + GRPO). V3: Qwen3-VL-8B-Instruct, zero-shot.

A key finding across all RL experiments is the stability–capability tradeoff. GRPO produced stable training curves but plateaued at 0.43 Dice, while PPO achieved higher peaks (0.60 Dice for V2) but suffered catastrophic collapse. V2′s sub-mask decomposition was observed only once (a transient spike to 2.8 sub-masks at PPO update 139, coinciding with peak Dice), confirming the strategy's value but also the inability of standard RL exploration to stabilize it. Oracle analysis shows decomposition yields up to 0.25 Dice improvement on spatially disconnected objects, representing substantial unrealized potential in the RL approaches.

Future Milestones

The remaining work focuses on distilling V3′s VLM-level performance into a compact, deployable model, exploring RL-based fine-tuning of VLMs to improve their use of segmentation and vision tools, and validating on additional domains.

Date	**Milestone**
Mar 28 – Apr 4	Scale VLM trajectory collection to 50–100 FSS-1000 classes
Apr 4 – Apr 8	Behavioral cloning distillation: train compact PolicyTransformer from VLM-generated demonstrations
Apr 8 – Apr 12	RL fine-tuning: GRPO on distilled policy and exploratory RL training of VLM to improve segmentation and vision tool use
Apr 12 – Apr 16	Full FSS-1000 test set evaluation and Kvasir-SEG [8] cross-domain evaluation
Apr 16 – Apr 20	Final ablation experiments and report writing
Apr 22 – Apr 27	Presentations

Table 2: Projected timeline through end of semester.

Conclusion

This project addresses automated prompt optimization for SAM in one-shot segmentation through four approaches spanning graph-based Q-learning, transformer policy networks, sub-mask decomposition, and zero-shot VLM guidance. Our experiments reveal that proxy rewards are dangerous (the Q-learning proxy was anti-correlated with true segmentation quality), that standard RL exploration cannot discover multi-step compositional strategies like sub-mask decomposition, and that behavioral cloning warmstarts are insufficient to prevent PPO collapse. The zero-shot VLM approach (V3) achieves the best results at 0.776 mean Dice with 1.3 clicks, outperforming all trained policies through chain-of-thought visual reasoning rather than trial-and-error exploration. The remaining work will distill V3′s capabilities into a compact, efficient model, investigate using RL to train a VLM to more effectively leverage segmentation and vision tools, and evaluate cross-domain generalization.

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